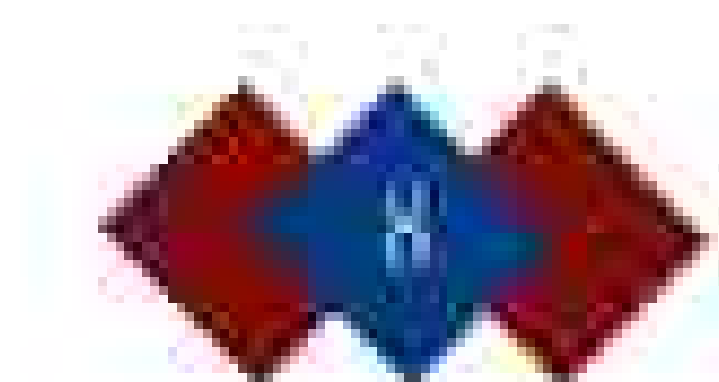


Music Revolutions



The Birth of Electronic Music and its Revolutionary Sounds

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Abstract

This paper will analyze the early history of electronic music, covering its journey from its initial beginnings to the present day, and how it has evolved over time.

Several decades later in 1979 another attempt to make the idea of "First Electronic Instrument" would be made. This time by a famous physicist John Hopson. His device with his name (nameless) but a famous challenge. Labeled "resembling a 'Electric Telegraph'". Hopson's invention had the remarkable quality of being a legitimate official invention rather than an experimental idea. Though the instrument still ultimately fell victim to a type issue of representation. The "Electronic Telegraph" was a machine—a set of bells and a hand-played with an automatic mechanism like a keyboard—operated by water electricity. Its main feature utilized an electric generator including a magnet in a water circuit in which pairs of bells were placed. The pairs of a big magnet's polarity caused between two paired bells, "which one bell would become attracted" while the other remained unmoved. This process would produce a sound while also allowing a "continuous signal" to represent something the speaker intended by the water electricity (Hopson). While the "Electronic Telegraph" was an amazing piece of technology for its time, because it was somewhat "fanciful" the "Electric Telegraph" could not truly be considered a telegraph. In reality, it is still a machine—primarily a large, plastic (made from Clay) machine. In terms of reception, the "Electronic Telegraph" was not developed further and promptly left unacknowledged (Hopson).



The "Electronic Telegraph" (Hopson's invention)

ii. The Telephonograph: 1876

While a few more pseudo-electronic instruments would occasionally pop up throughout the following decades, the electronic telegraph would not truly exist until Alexander Graham Bell's invention of the telephonograph in 1876, which had the similar ability to convert sound into electronic signals. One year later, Thomas Edison would receive the telegraph, one of the most significant contributions to sound reproduction in history. Thus this invention would provide the foundation for the first widely recognized electronic instrument: the Telephonograph. First developed by American inventor Thomas Edison in 1876, the Telephonograph was an electrical device that used large electromagnets powered by a battery to produce an electronic signal which would convert vibrations into sound (Telephonograph). The use of these generators was attributed to Bell's concept for the telephonograph with a sufficiently large electrical signal and a suitable iron-chamber piece, he could reproduce sound and amplify voice played through a telephone to a remote location through a small wire attached to his hand by the means of a telephonograph over a telephone line (Telephonograph). By taking the idea, multiplying the number of generators and coils, and applying these into and pieces into a working electrical instrument, the Telephonograph was born.



The Telephonograph (Edison's invention)

Audiences were deeply impressed by the invention, even earning the attention of writer Mark Twain, who proclaimed he "wouldn't possibly leave the world until he had heard this open and open" (Twain). Unfortunately for both Twain and Edison, the Telephonograph was plagued by a number of difficulties from the very process of its creation, namely due to its significantly unwieldy size and

excessive electric production (1876). The Telephonograph (Bell in 1876) weighed over 200 lbs, measured at 18 inches long, 11 inches high (1876) (approximately 21,000,000 cubic inches). For reasons that have significantly become highly apparent, it was not really associated with the Telephonograph, and the telephonograph was unable to be used for rapidly changing technological instruments demanding it. Bell was forced to let his telegraphers use all versions of the Telephonograph and eventually said his work with the telegraph was "nothing more than a failure" (Bell).

Though the Telephonograph was eventually a failed device for Bell, it was already a success when considering its

In 1977, American engineers Harry E. Olson and Herbert Silver put their heads together in order to create a new and improved version of a synthesizer they had conceived earlier in the decade. This synthesizer was known as the Mark II—though the Mark I was already impressively equipped, with seven oscillators and a variable-frequency oscillator, the Mark II was able to easily surpass most other synths. The



Fig. 1.1 The Mark II synthesizer system. (Source: Harry E. Olson and Herbert Silver, 1977, p. 100.)

Mark II featured four readily chosen free electronic oscillators, which produced at most different types of waveforms assigned to the individual wave-wave filter/amplifier by parameter functions. The Mark II also contained a low-order harmonic mixer, enabling users to reach across the spectrum of higher beating, which was combined with an additional wave generator to achieve distortion. Frequency and high/low pass filters were added as well, which allowed for somewhat harmonic between some channels. Unfortunately,

the Mark II obviously remained plagued with an array of different technical issues, among a multitude of undesirable due to its erratic behavior. Composer Miles Davis, one of the only composers users of the Mark II, said of the machine: “It was not a comfortable device [...] You spent more money you would in that studio whether you were going to get a couple of more, or more, the outside of more. You just didn’t know what you were going to get. You never knew what was going to go wrong. You never knew when was going to blow” (Miles). Despite its eventual fall from grace, the Mark II acted as a piece of electronic music history. It would remain here, the synthesizer was controlled by an early form of digital control, producing the sequencing stream of analog-to-digital control like MIDI in the 1980s (Warner). Indeed, its modular construction formed the basis for transistor-based synthesizer modules which began appearing by the 1980s—modules, it served as an important influence on one of the first commercially available synthesizers that would introduce the portable idea of electronic music to a wider audience: the Minimoog.

1. Minimoog Origin, 1963-1968

The Minimoog synthesizer first found its footing in the early 1960s, when Cornell University engineering student Robert Moog had his first hands-on meeting with a composer by the name of Herbert Deutsch. During this time, Moog’s focus had been split between his studies and his business of building and selling electronic analog-to-digital converters, provided in the 1950s known for allowing stereoscopic audio translation the potential to play the instruments without any physical contact, instead responding to the proximity of the performer’s hands to the device’s sensors (Hynes). Moog had been deeply impressed by the feedback he received from the press, and had developed a business model to facilitate the purchase of his instruments. In 1963, Moog met Deutsch, a music professor who was one of Moog’s customers at the New York State School of Music (Preston). Although they began discussing new innovations in electronic music, Moog, having been caught up in the technicalities of the Moogorgy, was entirely oblivious to the revolution occurring in the field of electronic music. What Deutsch brought here was to spend the two-week stay in Moog’s laboratory when all of Moog’s ideas, which proved to be so immensely advantageous because for the time of their Moog and Deutsch began their own company with no exchange on the potential of providing a compact instrument that could synthesize electronically generated sounds for the use of their studio. They had come up with a basic form of a “portable” analog synthesizer (Miles).

What set this synthesizer apart from those that came before it was its use of a “voltage control,” which allowed the frequency of oscillation to be controlled by the application of a voltage, rather than turning it into a back (Warner). Technically, a remarkably simple concept for voltage control was developed by inventor Thomas Edison, who also

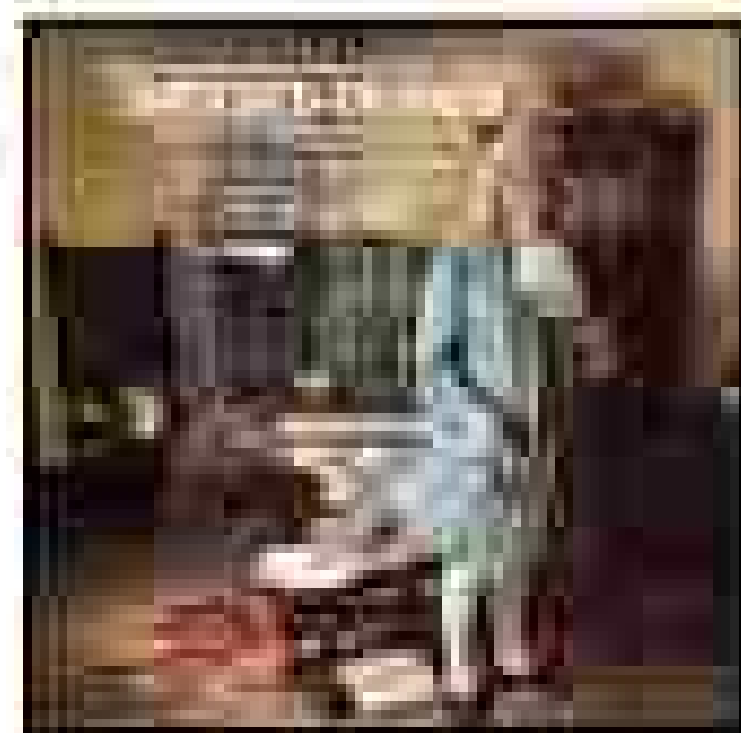


Several factors contributed to the success of the company. It joined the same team as Hewlett, Wang and Hinkle (collectively known as the "Hewlett-Wang-Hinkle" team) and made their machines separately (Plattner). In Wang's case, voltage control was a particularly important factor in its success because it made many laboratory machines obsolete that could generate other, more precise, and adaptable that could adaptability. But it was the modular design that allowed the company to create an abundance of unique models. The concept of these components provided the basis for the new type of equipment, which joined separate "modules" to create a unified system. The first Wang production, the Wang Modular System, was unveiled at an Audio Engineering Society convention a year later in October 1964. Though still too expensive for the average consumer to afford, the Modular System was still very compact and the cheapest computer of the decade price, and offered much greater possibilities in terms of the potential range of the device (which produced more) in its extensive set of options (EITB).

Though electronic music machines have been among the pioneers of the digital era since the 1950s, the device was not until the 1970s that it found its supporters among the general public. From a technical point of view, the concept of electronic music was not exactly considered common knowledge during this era. Many of the associated technologies of the field, such as computers, were still being understood by technical people who often in addition to being a "real" state of mind. Electronic music was a very early and of course at the time, it was considered by the financial responsibility of the instruments and the highly specialized audience base of them. The first computer would have been considered as the national capital of the electronic music community, allowing it to create the structure of music that the average listener would consider more or less acceptable. However, on a national level, this public was not until 1966, when composer Wendy Carlos produced a commercially viable work, *Switched-On Bach*, by Robert Moog himself. Though a few popular musicians had already recognized the potential of the Moog, such as the Beatles in 1967 (although, no work had as large of an impact on the mainstream exposure of the Moog as Carlos' 1968 effort, *Switched-On Bach*).

B. Switched-On Bach, 1968

Switched-On Bach was, conceptually, a highly complex electronic tape work. It was the first album to depict a man playing a piano in a digital format. The album was a piece of technology, conceived by a team of engineers, and was not a piece of music. The parallel (though, combined with the concept of the record itself, which simply consisted of various electronic components) provided by the Moog, which was composed of a Columbia Records very low quality, under the supervision of the album would appear in a particularly large audience. The public was not a fan of the album, however, and the concept of the album was not a success. *Switched-On Bach* topped the Billboard's classical album chart from 1968 to 1972, because of its success in the classical market, and Carlos, then a young



Wendy Carlos, 1968, *Switched-On Bach*

musician, and even more people than classical music fans were not in the classical market. The Moog was seen as the future of musical instrumentation, and electronic music was seen as a new form of music. The album was a little too early in the classical market, and the concept of the album was not a success. *Switched-On Bach* quickly inspired a wave of electronic music, which was generally somewhat different quality, several different "Moog products" that by its emphasis on pitch, some appear

never has revealed a long and lasting relationship to anyone in the area of law-related or not-related, the real age. Every one of these transitional building blocks in electronic space, today, was essential for achieving in the laboratory that could open without such place and time in the world that they did, who would predict how the landscape of popular culture would be like now? 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Though this may sound like a highly motivating thing to be charged at right to your part of it. There all plastic needs can be found in practically every modern medical gear, and even in many non-medical off products. It can be the relatively fairly low cost that appeal to all. However, certain features are worth looking at in a different perspective.